

Structural breaks in Box-Cox transforms of realized volatility: a model selection perspective

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Autoregressive (AR) models such as the heterogeneous autoregressive (HAR) model capture the linear footprint inherent in realized volatility. We draw upon the fact that the HAR model is a constrained AR model and cast the problem of estimating structural breaks in the autoregressive volatility dynamics as a model selection problem. A two-step Lasso-type procedure is used to consistently estimate the unknown number and timing of structural breaks. Empirically, we find the number of breaks to be heavily influenced by Box-Cox transformations applied to realized volatility series of eight stock market indices: For example, while we find breaks in the original series, no breaks are found in log-realized volatility, a measure often used in applied research, across a wide range of lag lengths. These Box-Cox transformations lead to different volatility processes with distinct autoregressive dynamics and affect the estimation of structural breaks. Importantly, the log-transformation considerably reduces the number of price jumps which might otherwise be selected as structural breaks.

Keywords: Realized volatility; Box-Cox transformation; Structural breaks; Group Lasso

JEL codes: C22, C52, G15

1. Introduction

One particularly important measure in finance is the volatility of exchange-traded assets. Denoting the variation of the price path over time, estimates and forecasts of volatility are, among other things, essential input factors for risk management and trading applications. Since financial markets change over time, an extensive literature has tested for and studied the implications of structural breaks in the latent volatility process (e.g. Starica and Granger 2005, Hillebrand 2005, Rapach and Strauss 2008, Smith 2008).

The investigation of such structural breaks in the latent volatility process is crucial for several reasons: Firstly, from a purely statistical point of view, structural breaks not only constitute nonlinearities that can be modeled but also introduce a subtle kind of nonstationarity. Given that the majority of endeavors related to time series modeling and forecasting rely on some form of stationarity assumption, any potential source of nonstationarity should be taken into account in an empirical analysis (e.g. see Casini and Perron 2019c, for an extensive overview of the related literature). Secondly, from an empirical finance perspective, the pronounced long-memory behavior generally observed for daily volatility time

series may also be partly attributable to neglected breaks in the underlying latent volatility process (for relevant references see, among others, Teverovsky and Taqqu 1997, Lobato and Savin 1998, Engle and Smith 1999, Parke 1999, Diebold and Inoue 2001, Gouriéroux and Jasiak 2001, Granger and Hyung 2004). On this note, see also Qu (2011) and McCloskey and Hill (2017) for the development of statistical tests and applications to volatility models. Thirdly, a thorough understanding of the implications of changing volatility dynamics and persistence is directly linked to time-varying risk premium theories along the lines of, among others, French *et al.* (1987) and Campbell and Hentschel (1992), which have been further addressed in several empirical studies (e.g. Bekaert and Wu 2000, Bollerslev *et al.* 2006). The volatility feedback effect relies on the pronounced persistence of volatility, which indicates that large news shocks (either positive or negative) affect not only current but also future volatility. To be more precise, an increase in volatility is assumed to lead to an increase in expected returns and a decrease in current stock prices, lowering volatility over time if the news shock is positive and further raising volatility if the news shock is negative (Bekaert and Wu 2000).

Given the availability of high-frequency data, realized measures have become essential in financial applications, which constitute ex-post measures of daily volatility (e.g. Andersen

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et al. 2001, Barndorff-Nielsen and Shephard 2002). Denoting by p_t the log-price of an asset and ignoring microstructure noise and price jumps, it is assumed that p_t evolves as $dp_t = \mu_t dt + \sigma_t dW_t$ in continuous time, where μ_t and σ_t are the instantaneous drift and volatility, respectively, and W_t is Brownian motion. It holds that $\text{plim}_{\delta \rightarrow 0} \sum_{t_i} (p_{t_{i+1}} - p_{t_i})^2 = \int_0^T \sigma_t^2 dt$, where $\delta = \sup\{t_{i+1} - t_i\}$. Thus, given an appropriate sampling frequency, an estimator for the integrated variance of trading day t , called realized variance, is given by $RV_t = \sum_{t_i} (p_{t_{i+1}} - p_{t_i})^2$, and its square root is the realized volatility.[†]

Empirical characteristics of RV_t and its square root include near log-normal unconditional distribution and long-memory behavior (e.g. Andersen *et al.* 2003). The latter is well-captured by the heterogeneous autoregressive (HAR) model of Corsi (2009) and its extensions. The initially introduced HAR model is a linear model incorporating weekly (w) and monthly (m) averages of daily (d) log RV_t (or, equivalently, of log-realized volatility):

$$\log RV_t^{(d)} = c + \beta^{(d)} \log RV_{t-1}^{(d)} + \beta^{(w)} \log RV_{t-1}^{(w)} + \beta^{(m)} \log RV_{t-1}^{(m)} + \varepsilon_t, \quad (1)$$

where $\log$ denotes the natural logarithm, $\log RV_{t-1}^{(w)} = \frac{1}{5} \sum_{i=1}^5 \log RV_{t-i}^{(d)}$, $\log RV_{t-1}^{(m)} = \frac{1}{22} \sum_{i=1}^{22} \log RV_{t-i}^{(d)}$, and $\{\varepsilon_t\}$ are zero mean innovations. The log-transformation is often applied to realized measures due to favorable finite sample properties (e.g. Gonçalves and Meddahi 2009, 2011). While the HAR model may not be the true data generating process (DGP) of RV_t , it captures the linear footprint inherent in realized measures, which explains its superior out-of-sample forecasting performance (Audrino and Knaus 2016).

Considering structural breaks in the HAR model, Fengler *et al.* (2015) find evidence for breaks in a linear and nonlinear specification of the model. Liu and Maheu (2008) and Dufays and Rombouts (2019), making use of Bayesian approaches, discover that breaks are present mostly in the variance of the logarithmic realized measure. Accordingly, other authors such as Bollerslev *et al.* (2016) and Bekierman and Manner (2018) intentionally allow the coefficients of the HAR model to vary across time based either on the estimated error variances or, more generally, driven by a latent Gaussian autoregressive process in order to provide a more flexible forecasting framework. Although an extensive literature exists that is concerned with the effects of (Box-Cox) transformations of RV_t on forecasting performance (see Taylor 2017, for a thorough analysis of the effects of Box-Cox transformations of realized variance on forecasting accuracy and comprehensive references to the related literature). The effects of Box-Cox transformations on the estimation of structural breaks in the autoregressive dynamics of the underlying latent volatility process have, thus far, not been thoroughly investigated.

To this end, we note that the above HAR model may also be written as a constrained AR process (Corsi 2009, Audrino

and Knaus 2016):

$$\log RV_t^{(d)} = \beta_0^* + \sum_{i=1}^{22} \beta_i^* \log RV_{t-i}^{(d)} + \varepsilon_t, \quad (2)$$

with the following restrictions being imposed on the AR coefficients:

$$\beta_i^* = \begin{cases} \beta^{(d)} + \frac{1}{5}\beta^{(w)} + \frac{1}{22}\beta^{(m)} & \text{for } i = 1, \\ \frac{1}{5}\beta^{(w)} + \frac{1}{22}\beta^{(m)} & \text{for } i = 2, \dots, 5, \\ \frac{1}{22}\beta^{(m)} & \text{for } i = 6, \dots, 22. \end{cases} \quad (3)$$

Building on the adaptive least absolute shrinkage and selection operator (Lasso) of Zou (2006) and extending the finding of Audrino and Knaus (2016) that a Lasso-determined AR model performs comparably to the HAR model, Audrino *et al.* (2015) find that the linear dynamics of such an AR model also change across time.

This paper adds to the previous literature in the following way: We cast the problem of structural break estimation in the autoregressive volatility dynamics as a model selection problem and apply the recently developed two-step group Lasso procedure of Chan *et al.* (2014), which permits the consistent estimation of the unknown number and timing of these breaks. Their procedure is related to a growing literature that investigates the performance of Lasso-type approaches for detecting and dating structural breaks (e.g. Ciuperca 2014, Rojas and Wahlberg 2014, Aue *et al.* 2014, Qian and Su 2016). In an application to realized volatility series of eight major stock market indices, two Box-Cox transformations of realized volatility are considered: The log-transformation often applied in the literature and realized precision, which is given as the inverse of realized volatility. While not used in applied research, the latter has been found to ensure favorable properties in simulation studies (Gonçalves and Meddahi 2011).

We discover that these transformations significantly change the autoregressive dynamics of the series and thus the number of detected structural breaks: For log-realized volatility, no breaks are found for most of the considered lag lengths, while the number of break-points for realized precision is generally greater than the number of break-points found for log-realized volatility but smaller than for untransformed realized volatility. Moreover, the timing of breaks for realized volatility and realized precision are similar. These findings remain unchanged qualitatively both for realized volatility estimators robust to microstructure noise and price jumps. In fact, the usage of a jump-robust realized volatility estimator does not necessarily decrease the number of detected structural breaks. Comparing the results of the considered Lasso-type structural break detection approach to the procedure proposed by Bai and Perron (1998, 2003), we find empirical support for our findings.

Lastly, we assess the effect of price jumps on the detection of structural breaks via the two-step group Lasso procedure in a Monte Carlo study based on a stochastic volatility jump diffusion model considered by, among others, Huang and Tauchen (2005). We find that the empirically observed benefit of the log-transformation to reduce the number of extreme values in the dependence on past values (i.e. to draw in outliers in the autoregressive structure) also has an impact on

[†] Note that this naive estimator is generally biased in the presence of microstructure noise or if p_t evolves as a jump-diffusion of the general form $dp_t = \mu_t dt + \sigma_t dW_t + dJ_t$, where J_t denotes a finite activity jump process (e.g. Audrino and Knaus 2016).

the detection of structural breaks in our simulation experiments. The transformation effectively reduces the effect of price jumps which might otherwise be falsely detected as structural breaks. Thus, from this perspective, it seems likely that the results obtained for the log-realized volatility conform better to reality and that there are likely zero or very few structural breaks in the dynamics of most of the realized volatility series considered in our empirical application. This finding extends the potential benefits related to the application of the log-transformation in empirical applications involving realized volatility estimates.

Overall, the significant difference in the number of estimated structural breaks in our empirical application is striking. Thus, when asking *if* there are structural breaks in realized volatility, we may also ask how the approach we choose to detect them is affected by the transformation of the series and, ultimately, which breaks may then be deemed true structural breaks. Since we test for changes in the persistence in the volatility series, our results could have implications for studies addressing time-varying risk premia (for example Bollerslev *et al.* 2011, make use of untransformed realized volatility series).

The remainder of the paper is structured as follows: Section 2 briefly explains the two-step break-point estimation procedure of Chan *et al.* (2014). The data set used in the empirical analysis and some empirical characteristics of the volatility series are described in Section 3. Next, Section 4 discusses the results of the Lasso-type break-point estimation in the autoregressive volatility dynamics and compares them to the results obtained by the procedure of Bai and Perron (1998, 2003). Section 5 offers some simulation evidence on the performance of the two-step estimation procedure in the presence of jumps. Lastly, Section 6 concludes.

2. Two-step structural break estimation

2.1. First step group Lasso estimation

Segmented locally stationary models (e.g. Casini 2019) are an easy to interpret means for structural break detection since break-points are indicated through changes in one or more parameters of the model. Following the general outline in Chan *et al.* (2014), we consider a structural break AR process with $(m+1)$ regimes to model the constrained AR process in equation (2), which is derived from the simple linear HAR model of realized volatility dynamics:

$$Y_t = \sum_{j=1}^{m+1} \left[\beta_j' Y_{t-1} + \sigma(Y_{t-1}, \dots, Y_{t-q}) \varepsilon_t \right] \mathbb{1}_{\{t_{j-1} \leq t < t_j\}},$$

$$t = 1, 2, \dots, n, \quad (4)$$

where $Y_{t-1} = (1, Y_{t-1}, \dots, Y_{t-p})'$, $\beta_j = (\beta_{j0}, \beta_{j1}, \dots, \beta_{jp})' \in \mathbb{R}^{p+1}$, $1 = t_0 < t_1 < \dots < t_{m+1} = n+1$, and $\sigma(\cdot)$ denoting a measurable function on $\mathbb{R}^q$, allowing for possibly conditionally heteroscedastic errors. The $\{\varepsilon_t\}$ are zero mean and unit variance white noise. Break-points are collected in the index set $\mathbf{t} = \{t_1, \dots, t_m\}$, i.e. the parameter β_j

changes to β_{j+1} at time t_j . We can rewrite the model in equation (4) as a high dimensional regression model by introducing the following notation: $Y_n^0 = (Y_1, Y_2, \dots, Y_n)'$, $\eta(n) = (\sigma_1 \varepsilon_1, \sigma_2 \varepsilon_2, \dots, \sigma_n \varepsilon_n)'$, with $\sigma_t = \sigma(Y_{t-1}, \dots, Y_{t-q})$, $\theta(n) = (\theta_1, \theta_2, \dots, \theta_n)'$, and X_n denoting an $(n \times np)$ -matrix:

$$X_n = \begin{pmatrix} Y_0' & 0 & 0 & \dots & 0 \\ Y_1' & Y_1' & 0 & \dots & 0 \\ Y_2' & Y_2' & Y_2' & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Y_{n-1}' & Y_{n-1}' & Y_{n-1}' & \dots & Y_{n-1}' \end{pmatrix}, \quad (5)$$

where $Y_k' = (Y_k, Y_{k-1}, \dots, Y_{k-p+1})$. Letting $\theta(n)$ denote the vector of all potential parameter changes, we set $\theta_1 = \beta_1$ for the first parameter and

$$\theta_i = \begin{cases} \beta_{j+1} - \beta_j & \text{when } i = t_j \text{ and } t_j \text{ is a break-point,} \\ 0 & \text{otherwise,} \end{cases} \quad (6)$$

for $i = 2, \dots, n$. The model in equation (4) can now be expressed as:

$$Y_n^0 = X_n \theta(n) + \eta(n). \quad (7)$$

A sparse solution for $\theta(n)$ is given by the group Lasso estimator of Yuan and Lin (2006):

$$\hat{\theta}(n) = \arg \min_{\theta(n)} \frac{1}{n} \|Y_n^0 - X_n \theta(n)\|^2 + \lambda_n \sum_{i=1}^n \|\theta_i\|, \quad (8)$$

where λ_n denotes a tuning parameter and $\|\cdot\|$ the Euclidean norm. The m break-points can be estimated by identifying the nonzero $\hat{\theta}_i$ since $\hat{\theta}_i \neq 0$ for an $i \geq 2$ implies a change in the AR parameter $\hat{\beta}_i$. Estimation is carried out with the group-LARS algorithm of Yuan and Lin (2006) where the upper bound for the number of breaks k has to be specified beforehand. See also the supplementary material of Chan *et al.* (2014) for detailed information about the implementation of the algorithm. Note also that, given the above notation, the AR order p is assumed to be constant across regimes, which may be relaxed in practice by first determining the breaks with a large enough value for p and then the appropriate lag order in each segment (see Remark 2.6 in Chan *et al.* 2014).

2.2. Second step exhaustive search

Chan *et al.* (2014) prove consistency of the AR parameter estimates (in terms of prediction error) and the break-points when m is unknown. However, in such a setting, group Lasso overestimates the true number of break-points, which is why active breaks are selected in a second step. We follow Chan *et al.* (2014) and Behrendt and Schweikert (2020) by defining an information criterion (IC) based on the minimum description length (MDL) principle of Davis *et al.* (2006). In the following, let $\tilde{\beta}_j$ denote the least squares estimator of the coefficient vector in segment j and $S_n(t_{j-1}, t_j) = \sum_{t=t_{j-1}}^{t_j-1} (Y_t - \tilde{\beta}_j' Y_{t-1})^2$ the residual sum of squares for segment j . In general, the

IC($m, \mathbf{t}$) may then be expressed as the sum of a goodness-of-fit measure and a penalty term:

$$\text{IC}(m, \mathbf{t}) = S_n(t_1, \dots, t_m) + m \cdot \omega_n, \quad (9)$$

where $S_n(t_1, \dots, t_m) = \sum_{j=1}^{m+1} S_n(t_{j-1}, t_j)$ is the residual sum of squares and ω_n the penalty term. In our case, ω_n is now given by the MDL of Davis *et al.* (2006). Collecting the estimated change points in the index set $\mathcal{A}_n = \{i \geq 2 : \hat{\theta}_i \neq 0\}$, we can estimate the number and timing of structural breaks by solving:

$$\begin{aligned} \arg \min_{m, \mathbf{t}} \text{IC}(m, \mathbf{t}), \quad m \in \{0, 1, \dots, |\mathcal{A}_n|\}, \\ \mathbf{t} \in \{t_1, \dots, t_m\} \subset \mathcal{A}_n. \end{aligned} \quad (10)$$

Thus, active breaks are selected from the set of break-point candidates, as estimated in the first step, via minimization of the above IC. This minimization is repeated in a backward elimination algorithm that purges the set of superfluous breaks, i.e. breaks corresponding to the largest reductions of the IC if excluded from the set. Following the outline in Chan *et al.* (2014), the algorithm works as follows:

- (1) Initialize the algorithm by setting $k = |\mathcal{A}_n|$, $\mathbf{t}_k = \{t_{k,1}, \dots, t_{k,k}\} = \mathcal{A}_n$, and $\Phi_k^* = \text{IC}(k, \mathcal{A}_n)$.
- (2) Compute $\Phi_{k,i} = \text{IC}(k-1, \mathbf{t}_k \setminus \{t_{k,i}\})$ for each $i \in \{1, \dots, k\}$ and set $\Phi_{k-1}^* = \min_i \Phi_{k,i}$.
- (3) Check the following conditions:
 - 1 If $\Phi_{k-1}^* > \Phi_k^*$, then the timing of structural breaks is given by $\mathcal{A}_n^* = \mathbf{t}_k$.
 - 2 If $\Phi_{k-1}^* \leq \Phi_k^*$ and $k = 1$, then $\mathcal{A}_n^* = \emptyset$, which indicates that there is no structural break in the given time series.
 - 3 If $\Phi_{k-1}^* \leq \Phi_k^*$ and $k > 1$, then proceed by setting $j = \arg \min_i \Phi_{k,i}$, $\mathbf{t}_{k-1} = \mathbf{t}_k \setminus \{t_{k-1,j}\}$, and $k = k-1$. Return to the second step of the algorithm.

Importantly, consistency of the number and timing of structural breaks can also be proven for the estimate $\mathcal{A}_n^*$, which we obtain from the above backward elimination algorithm (see Theorem 2.5 in Chan *et al.* 2014). In the following, we apply this two-step estimation procedure to several (transformed) realized volatility series.

3. Data

3.1. Jump and microstructure noise robust volatility estimators

In the empirical application below, we focus on the jump robust realized variance estimator MedRV of Andersen *et al.* (2012) and the realized kernel estimator of Barndorff-Nielsen *et al.* (2008), which is robust to microstructure noise induced by, for example, lack of liquidity, price discreteness, and the bid-ask bounce. The former constitutes an especially relevant estimator since the robustness against jumps is a helpful remedy against the potential detection of spurious breaks induced by these jumps. We compare results to realized volatility series based on the realized kernel variance estimator in order to investigate to what extent results might

be driven by jumps in the realized variance process, while, at the same time, mitigating the influence of microstructure noise.

To this end, the log-return $r_{i,t}$ of an asset at time i on trading day t , having a total of T intraday observations, is calculated as $r_{i,t} = p_{i,t} - p_{i-1,t}$, where $p_{i,t}$ is the log-price sampled at equidistant intraday time intervals. The MedRV estimate for trading day t is given by:

$$\text{MedRV}_t = \frac{\pi}{6 - 4\sqrt{3} + \pi} \left(\frac{T}{T-2} \right) \sum_{h=2}^{T-1} \text{med}(|r_{i-1,t}|, |r_{i,t}|, |r_{i+1,t}|)^2, \quad (11)$$

where $\pi/(6 - 4\sqrt{3} + \pi)$ and $T/(T-2)$ are scaling factors and $\text{med}(\cdot)$ is taking the median value over the three adjacent intraday returns. On the other hand, the realized kernel variance estimate for trading day t is given by:

$$\text{KernelRV}_t = \sum_{h=-H}^H k\left(\frac{h}{H+1}\right) \eta_h, \quad (12)$$

where $k(\cdot)$ is the Parzen kernel function and $\eta_h = \sum_{i=|h|+1}^T r_{i,t} \cdot r_{i-|h|,t}$. See Barndorff-Nielsen *et al.* (2009) for a discussion of the optimal choice for the bandwidth H . MedRV and realized kernel variance time series of eight major stock market indices are obtained from the Oxford-Man Institute of Quantitative Finance's Realized Library (Heber *et al.* 2009). For a further description of the data set, data sampling and aggregation, and the considered estimators, we refer to the information and related literature that can be accessed via the Realized Library.[†] The stock market indices considered here are: Dow Jones Industrial Average (DJIA), Cotation Assistée en Continu (CAC 40), Financial Times Stock Exchange Index (FTSE 100), German Stock Index (DAX), Hang Seng Index (Hang Seng), Nikkei 225, Standard & Poor's 500 (S&P 500), and Shanghai Stock Exchange Composite Index (Shanghai Composite). Overall, the data cover the period from January 3, 2000, to December 31, 2018. Although we discuss results only for the realized volatility measures based on the MedRV in equation (11) and the realized kernel in equation (12), our results remain unchanged qualitatively for other realized measures that can be obtained from the Realized Library. These results are available upon request. Moreover, we have also considered realized volatility series based on MedRV for a sample of liquid stocks as a robustness check. The data are essentially the same as in Audrino *et al.* (2020), and we refer the interested reader to their paper for summary statistics and further information on the sample of stocks.

3.2. Empirical characteristics of (transformed) realized volatility

Three volatility series are investigated for the presence of structural breaks: Realized volatility, given as the square root of equations (11) and (12), the logarithm of daily realized volatility, a measure predominantly used in the literature

[†] See <https://realized.oxford-man.ox.ac.uk/data>.

Table 1. Descriptive statistics for realized volatility series based on MedRV estimator.

Index	Obs.	Mean	St. Dev.	Min	Max	Skew.	Kurt.
Panel A: realized volatility							
DJIA	4762	0.0053	0.0039	0.0007	0.0542	3.4707	25.1479
CAC 40	4842	0.0067	0.0041	0.0003	0.0483	2.5592	15.0957
FTSE 100	4785	0.0053	0.0033	0.0003	0.0385	2.9585	18.4880
DAX	4818	0.0070	0.0045	0.0011	0.0550	2.3770	12.4890
Hang Seng	4658	0.0039	0.0022	0.0006	0.0358	3.1802	23.8530
Nikkei 225	4627	0.0046	0.0027	0.0007	0.0339	2.7028	18.0635
S&P 500	4768	0.0055	0.0042	0.0006	0.0537	3.3644	23.4395
Shanghai Composite	4583	0.0055	0.0038	0.0000	0.0462	2.3014	11.8326
Panel B: log-realized volatility							
DJIA	4762	− 5.4211	0.5843	− 7.2612	− 2.9153	0.3766	3.2593
CAC 40	4842	− 5.1545	0.5363	− 8.1837	− 3.0295	0.0946	3.8487
FTSE 100	4785	− 5.3832	0.5003	− 8.1362	− 3.2572	0.5071	3.6815
DAX	4818	− 5.1191	0.5477	− 6.7966	− 2.9002	0.3404	2.9856
Hang Seng	4658	− 5.6771	0.4732	− 7.4230	− 3.3303	0.3830	3.7104
Nikkei 225	4627	− 5.5042	0.5067	− 7.2088	− 3.3844	0.1511	3.2107
S&P 500	4768	− 5.4002	0.6058	− 7.3671	− 2.9238	0.3153	3.2058
Shanghai Composite	4583	− 5.3815	0.5989	− 11.5934	− 3.0748	0.0197	5.4142
Panel C: realized precision							
DJIA	4762	265.1809	151.0810	18.4543	1423.9487	1.3949	6.6194
CAC 40	4842	200.3974	136.1056	20.6871	3581.9516	7.8469	139.7331
FTSE 100	4785	244.6955	125.4418	25.9763	3415.9591	4.9983	103.3904
DAX	4818	192.4526	101.9409	18.1774	894.8396	1.1492	5.0616
Hang Seng	4658	324.9328	153.1509	27.9458	1674.0730	1.4756	8.0457
Nikkei 225	4627	278.5905	145.3564	29.4999	1351.3206	1.4587	6.3685
S&P 500	4768	263.2332	159.2017	18.6121	1582.9970	1.6407	8.0490
Shanghai Composite	4583	280.0233	1604.6941	21.6454	108 376.3109	66.6988	4492.8764

Notes: This table provides summary statistics for the (transformed) realized volatility time series based on the MedRV estimator of Andersen *et al.* (2012). In total, realized volatility time series of eight major stock market indices are considered for the time period from January 3, 2000, to December 31, 2018. Shown are the number of observations and sample statistics such as mean, standard deviation, minimum, maximum, skewness, and kurtosis. Panel A depicts the descriptive statistics for realized volatility, calculated as the square root of equation (11), Panel B for the corresponding log-realized volatility, and Panel C for realized precision, denoting the inverse of realized volatility.

(for example, Jung and Maderitsch 2014, use this measure to investigate volatility spillovers), and realized precision, denoting the reciprocal of realized volatility. While the latter is not a measure regularly used in applied research, Gonçalves and Meddahi (2011) derive it from a Box-Cox transformation of realized volatility and find it favorable if one aims to control the coverage probability of integrated variance by 95% confidence intervals. Here, it serves as another (nonlinear) transformation of realized volatility that is used in our comparison of structural breaks and linear dynamics given its good approximation to the true integrated variance.

For the empirical analysis, we remove all trading days with $MedRV_t = 0$, which would prohibit the estimation of structural breaks for the transformed realized volatility series. However, for the Shanghai Composite, we still observe a very small value for $\sqrt{MedRV_t}$ on February 11, 2003, which significantly affects our results reported in the next section. The realized volatility series based on the realized kernel variance estimator do not contain such a case. Tables 1 and 2 provide descriptive statistics for all three series across the eight stock market indices. The log-transformation significantly reduces the skewness and kurtosis of the original realized volatility data. In fact, plots of kernel densities for the log-realized volatility series together with superimposed normal densities, reveal the pronounced empirical closeness to the normal distribution. Nonetheless, considering the log-realized volatility based on MedRV, the kurtosis for the Shanghai Composite is larger than for all other indices, reflecting the

outlier mentioned above. The autocorrelation functions for the three series differ slightly, but all feature the well-known long-memory behavior.[†] Note, however, that without taking structural breaks into account, these may not mirror the true dynamics of the series.

Moreover, the log-transformation considerably reduces the number of extreme values in the dependence on past values. As an example, figure 1 depicts scatter plots of the original and the lagged series for the S&P 500 based on MedRV. While the correlation for log-realized volatility is similar to the one for the realized volatility series, the data are more concentrated. Thus, the number of outliers that are visible for the realized volatility series are reduced significantly in the case of log-realized volatility. On the other hand, extreme values are still present in the case of realized precision.

4. Structural breaks in realized volatility dynamics

4.1. Structural breaks detected by the two-step group Lasso procedure

We follow Chan *et al.* (2014) and set the upper bound for the number of breaks in our application to $k = 10$. However, results do not change significantly for different choices

[†] Both the kernel density plots and the autocorrelation function plots are available upon request.

Table 2. Descriptive statistics for realized volatility series based on realized kernel estimator.

Index	Obs.	Mean	St. Dev.	Min	Max	Skew.	Kurt.
Panel A: realized volatility							
DJIA	4762	0.0084	0.0060	0.0010	0.0823	3.1165	20.1606
CAC 40	4842	0.0098	0.0062	0.0014	0.0694	2.5216	14.5776
FTSE 100	4785	0.0092	0.0062	0.0015	0.0941	3.5888	28.8668
DAX	4818	0.0106	0.0069	0.0013	0.0872	2.7504	17.0276
Hang Seng	4658	0.0086	0.0052	0.0013	0.0824	3.4280	26.7428
Nikkei 225	4627	0.0088	0.0061	0.0010	0.0906	4.3075	39.5018
S&P 500	4768	0.0082	0.0061	0.0009	0.0715	3.0993	19.4520
Shanghai Composite	4583	0.0114	0.0070	0.0021	0.0692	2.1918	10.4668
Panel B: log-realized volatility							
DJIA	4762	−4.9602	0.5862	−6.8781	−2.4969	0.2224	3.3292
CAC 40	4842	−4.7872	0.5544	−6.5636	−2.6673	0.1520	3.1124
FTSE 100	4785	−4.8385	0.5300	−6.4910	−2.3630	0.4340	3.4270
DAX	4818	−4.7047	0.5486	−6.6086	−2.4395	0.2566	3.2198
Hang Seng	4658	−4.8927	0.4927	−6.6106	−2.4958	0.4046	3.6597
Nikkei 225	4627	−4.8911	0.5457	−6.8654	−2.4016	0.2536	3.6052
S&P 500	4768	−4.9959	0.6085	−6.9605	−2.6385	0.2298	3.1788
Shanghai Composite	4583	−4.6243	0.5282	−6.1747	−2.6713	0.3002	3.0510
Panel C: realized precision							
DJIA	4762	168.3280	102.3253	12.1448	970.8106	1.8159	8.6792
CAC 40	4842	139.3385	80.0186	14.4006	708.8050	1.6344	7.4109
FTSE 100	4785	143.9421	73.6273	10.6229	659.1837	1.2412	5.9942
DAX	4818	127.5973	70.8725	11.4675	741.4742	1.5727	7.6025
Hang Seng	4658	149.5234	72.8319	12.1315	742.9144	1.4342	7.4308
Nikkei 225	4627	153.6760	86.3685	11.0406	958.5651	1.6943	8.3037
S&P 500	4768	176.4769	109.9547	13.9919	1054.1414	1.7150	7.9084
Shanghai Composite	4583	116.3606	60.5936	14.4585	480.4282	1.2775	5.7169

Notes: This table provides summary statistics for the (transformed) realized volatility time series based on the realized kernel estimator of Barndorff-Nielsen *et al.* (2008). In total, realized volatility time series of eight major stock market indices are considered for the time period from January 3, 2000, to December 31, 2018. Shown are the number of observations and sample statistics such as mean, standard deviation, minimum, maximum, skewness, and kurtosis. Panel A depicts the descriptive statistics for realized volatility, calculated as the square root of equation (12), Panel B for the corresponding log-realized volatility, and Panel C for realized precision, denoting the inverse of realized volatility.

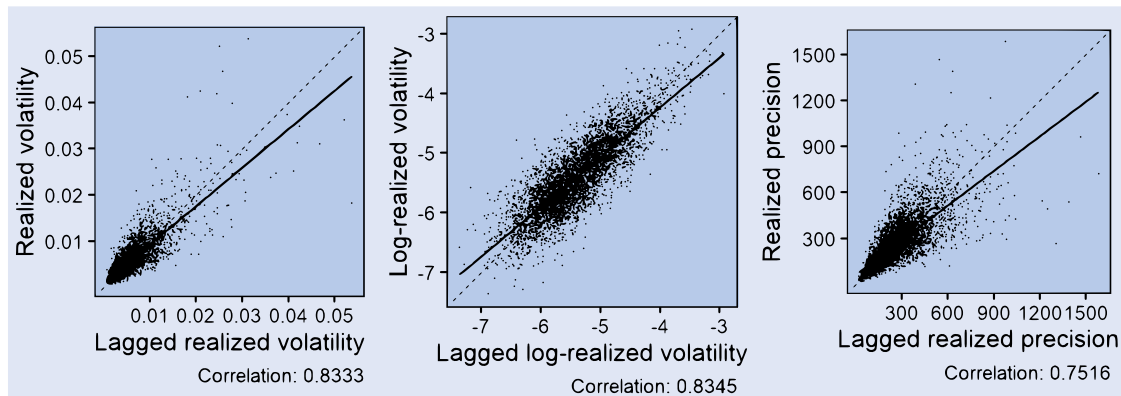


Figure 1. Scatter plots of volatility series and their first lags for the S&P 500 (2000–2018), MedRV estimator.

The plots depict scatter plots for realized volatility (left), log-realized volatility (middle), and realized precision (right), based on the MedRV estimator of Andersen *et al.* (2012), with the first lag of the respective series for the S&P 500 from January 3, 2000, to December 31, 2018. Dashed lines indicate the case of perfect correlation and solid lines are a linear fit to the data. Gray shaded areas around the regression lines depict 95% confidence intervals. The respective correlation is given below each scatter plot.

of k . Tables 3 (measures based on MedRV) and 4 (measures based on the realized kernel) depict the results of the two-step estimation approach for all stock market indices and the three time series considered across a variety of reasonable choices for the lag structure, ranging from $p = 2$ to $p = 22$, which are active in each detected regime.[†] As a comparison,

in an application to squared returns of the S&P 500, Chan *et al.* (2014) use a lag length of $p = 12$. We find that, depending on the stock market index, results are reasonably robust for different choices of the lag structure but mostly for lag lengths ranging from $p = 8$ to $p = 18$. It is striking that while the dynamics of realized volatility exhibit structural breaks

[†] For some realized volatility series, the group-LARS algorithm fails to converge for $p > 20$, which is denoted in the tables by ‘–’. To be

more precise, in these cases, a nearly singular matrix is encountered when computing the decent direction in the group-LARS algorithm.

Table 3. Number of structural breaks for realized volatility series based on MedRV estimator.

Index	AR(2)	AR(4)	AR(6)	AR(8)	AR(10)	AR(12)	AR(14)	AR(16)	AR(18)	AR(20)	AR(22)
Panel A: realized volatility											
DJIA	3	2	2	2	3	3	3	4	4	2	2
CAC 40	5	4	5	4	5	4	4	4	4	3	3
FTSE 100	3	4	3	3	4	4	4	4	4	3	3
DAX	6	5	5	5	4	2	3	3	3	2	3
Hang Seng	4	4	4	4	3	2	3	2	3	2	2
Nikkei 225	4	4	4	4	3	3	4	5	4	2	2
S&P 500	5	3	2	2	3	3	3	4	4	3	4
Shanghai Composite	2	3	3	3	3	3	3	3	3	3	–
Panel B: log-realized volatility											
DJIA	1	1	1	0	0	0	0	0	0	0	0
CAC 40	1	0	0	0	0	0	0	0	0	0	0
FTSE 100	3	0	0	0	0	0	0	0	0	0	0
DAX	1	0	0	0	0	0	0	0	0	0	0
Hang Seng	2	2	0	0	0	0	0	0	0	0	0
Nikkei 225	1	0	0	0	0	0	0	0	0	0	0
S&P 500	1	1	1	0	0	0	0	0	0	2	4
Shanghai Composite	2	1	1	1	1	1	1	1	1	1	2
Panel C: realized precision											
DJIA	2	2	2	2	1	2	2	2	2	2	1
CAC 40	4	4	3	2	2	1	1	1	1	2	3
FTSE 100	4	3	1	3	1	1	1	1	1	2	1
DAX	3	3	1	1	1	1	1	1	1	2	3
Hang Seng	4	1	2	2	2	2	2	0	0	0	0
Nikkei 225	1	1	2	1	1	1	1	1	1	3	–
S&P 500	2	2	2	2	2	1	1	2	2	3	2
Shanghai Composite	6	5	5	4	3	3	3	3	3	3	–

Notes: This table lists the estimated number of break-points by the procedure of Chan *et al.* (2014) for all eight stock market indices and the realized volatility (Panel A), log-realized volatility (Panel B), and realized precision (Panel C) time series based on the MedRV estimator of Andersen *et al.* (2012) across different AR specifications. The upper bound for the number of breaks is set to $k = 10$.

Table 4. Number of structural breaks for realized volatility series based on realized kernel estimator.

Index	AR(2)	AR(4)	AR(6)	AR(8)	AR(10)	AR(12)	AR(14)	AR(16)	AR(18)	AR(20)	AR(22)
Panel A: realized volatility											
DJIA	5	4	4	4	5	3	3	3	3	3	3
CAC 40	4	4	4	4	4	4	4	3	3	3	3
FTSE 100	4	4	4	4	4	4	4	4	4	4	3
DAX	6	6	4	3	2	4	5	5	3	2	2
Hang Seng	4	3	2	3	3	3	3	3	3	2	2
Nikkei 225	4	3	3	4	3	3	3	3	3	2	2
S&P 500	5	4	4	4	4	4	4	3	3	3	4
Shanghai Composite	3	3	3	3	3	3	3	3	4	1	1
Panel B: log-realized volatility											
DJIA	4	2	0	0	0	0	0	0	0	0	0
CAC 40	1	0	0	0	0	0	0	0	0	0	0
FTSE 100	3	0	0	0	0	0	0	0	0	0	0
DAX	1	0	0	0	0	0	0	0	0	0	3
Hang Seng	1	0	0	0	0	0	0	0	0	1	–
Nikkei 225	1	1	0	0	0	0	0	0	0	2	0
S&P 500	2	1	1	1	0	0	0	0	0	2	0
Shanghai Composite	3	2	0	0	0	0	0	0	0	1	–
Panel C: realized precision											
DJIA	3	2	2	2	2	2	2	2	2	2	–
CAC 40	3	3	2	2	1	3	1	3	1	1	2
FTSE 100	2	1	1	0	0	3	2	2	0	2	–
DAX	1	1	1	1	1	1	1	1	1	2	2
Hang Seng	2	2	1	1	1	1	1	1	1	2	–
Nikkei 225	2	4	3	3	1	2	1	1	1	2	4
S&P 500	2	2	2	2	2	2	2	1	1	2	2
Shanghai Composite	1	1	1	1	1	1	1	1	1	1	3

Notes: This table lists the estimated number of break-points by the procedure of Chan *et al.* (2014) for all eight stock market indices and the realized volatility (Panel A), log-realized volatility (Panel B), and realized precision (Panel C) time series based on the realized kernel estimator of Barndorff-Nielsen *et al.* (2008) across different AR specifications. The upper bound for the number of breaks is set to $k = 10$.

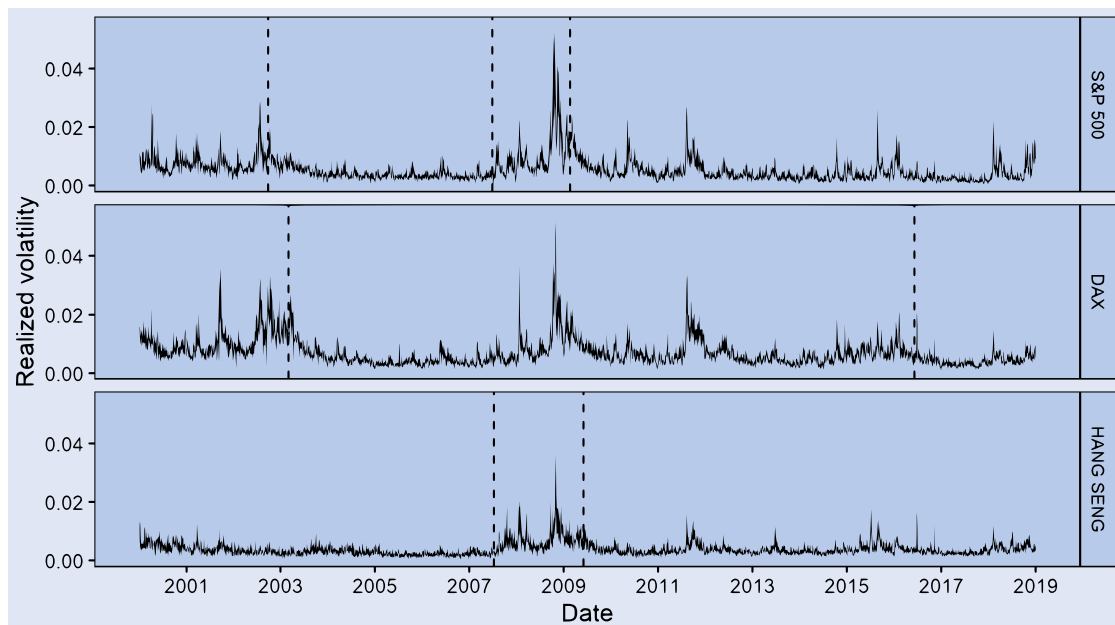


Figure 2. Realized volatility and break-points for three stock market indices (2000–2018), MedRV estimator.

This figure shows plots of realized volatility based on the MedRV estimator of Andersen *et al.* (2012) for the three stock market indices S&P 500 (first panel), DAX (second panel), and Hang Seng (third panel) from January 3, 2000, to December 31, 2018. Daily volatility estimates are based on the estimator in equation (11). Break-points are estimated using the two-step estimation procedure of Chan *et al.* (2014) with $k = 10$ and $p = 12$ and are highlighted by vertical dashed lines.

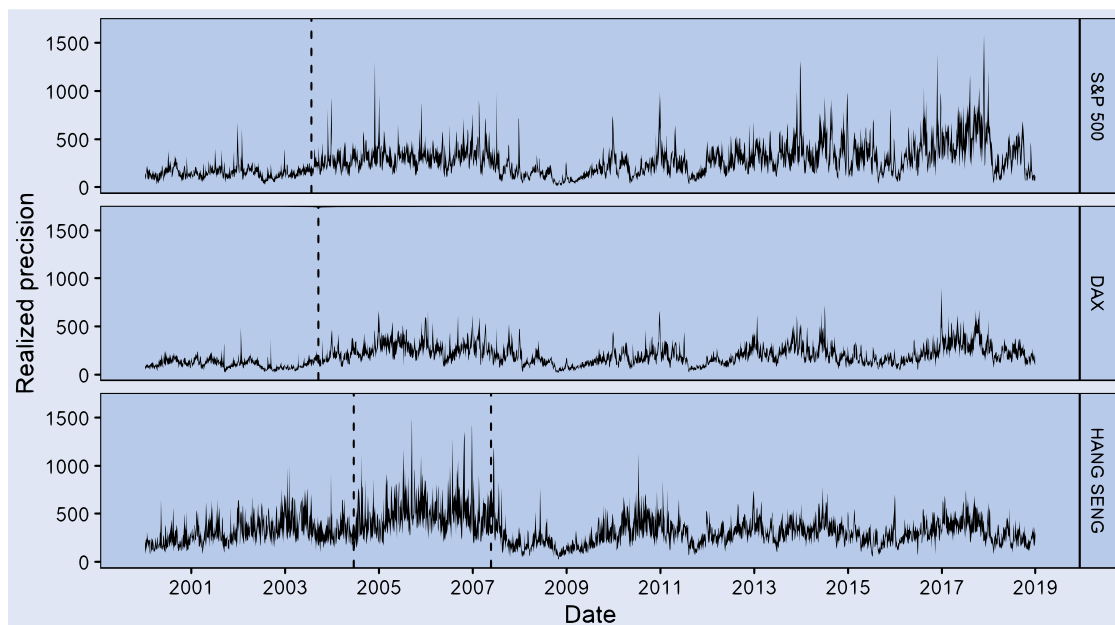


Figure 3. Realized precision and break-points for three stock market indices (2000–2018), MedRV estimator.

This figure shows plots of realized precision based on the MedRV estimator of Andersen *et al.* (2012) for the three stock market indices S&P 500 (first panel), DAX (second panel), and Hang Seng (third panel) from January 3, 2000, to December 31, 2018. Daily volatility estimates are based on the estimator in equation (11). Break-points are estimated using the two-step estimation procedure of Chan *et al.* (2014) with $k = 10$ and $p = 12$ and are highlighted by vertical dashed lines.

for all indices, no breaks are found for log-realized volatility across a wide range of lag lengths. In fact, only in the case of an AR(2) does the procedure of Chan *et al.* (2014) detect breaks in log-realized volatility for all indices. For realized precision, we mostly find a positive number of structural breaks across indices, while the number of detected breaks is often smaller compared to the untransformed realized volatility series. However, in many instances, the estimated breaks

are dated relatively close to the breaks detected for realized volatility. Figures 2–5, considering realized volatility and realized precision, offer a visual comparison of this empirical finding for three indices and measures based on MedRV and the realized kernel estimator. Moreover, the parameter changes are not restricted to changes in the intercept but involve the whole coefficient vector, i.e. we clearly detect changes in the autoregressive dynamics. Lastly, (transformed)

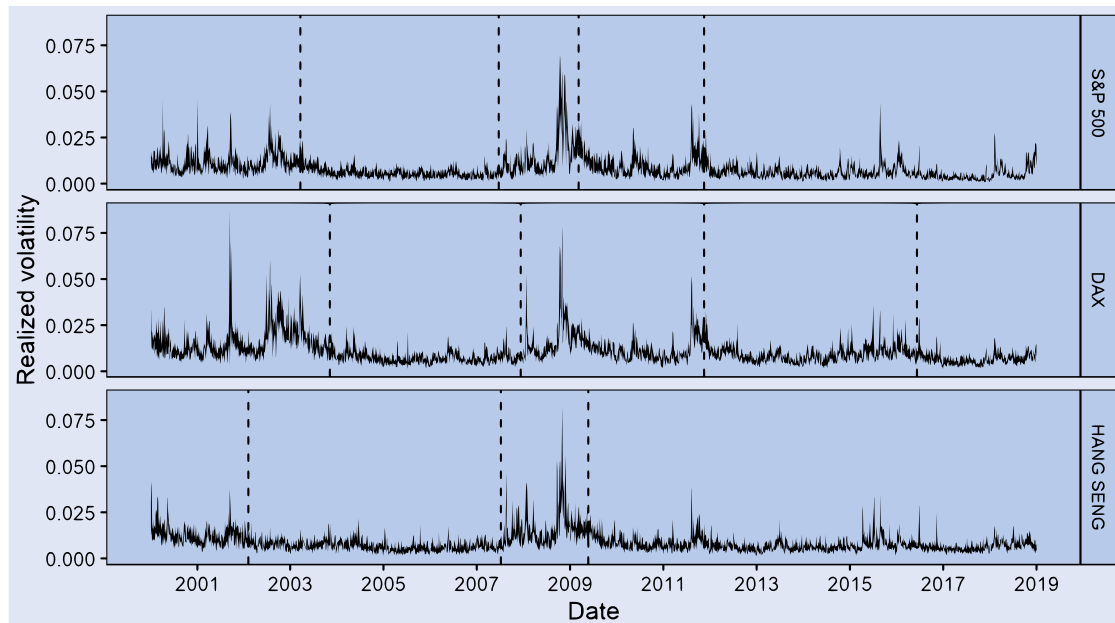


Figure 4. Realized volatility and break-points for three stock market indices (2000–2018), realized kernel estimator. This figure shows plots of realized volatility based on the realized kernel estimator of Barndorff-Nielsen *et al.* (2008) for the three stock market indices S&P 500 (first panel), DAX (second panel), and Hang Seng (third panel) from January 3, 2000, to December 31, 2018. Daily volatility estimates are based on the realized kernel in equation (12). Break-points are estimated using the two-step estimation procedure of Chan *et al.* (2014) with $k = 10$ and $p = 12$ and are highlighted by vertical dashed lines.

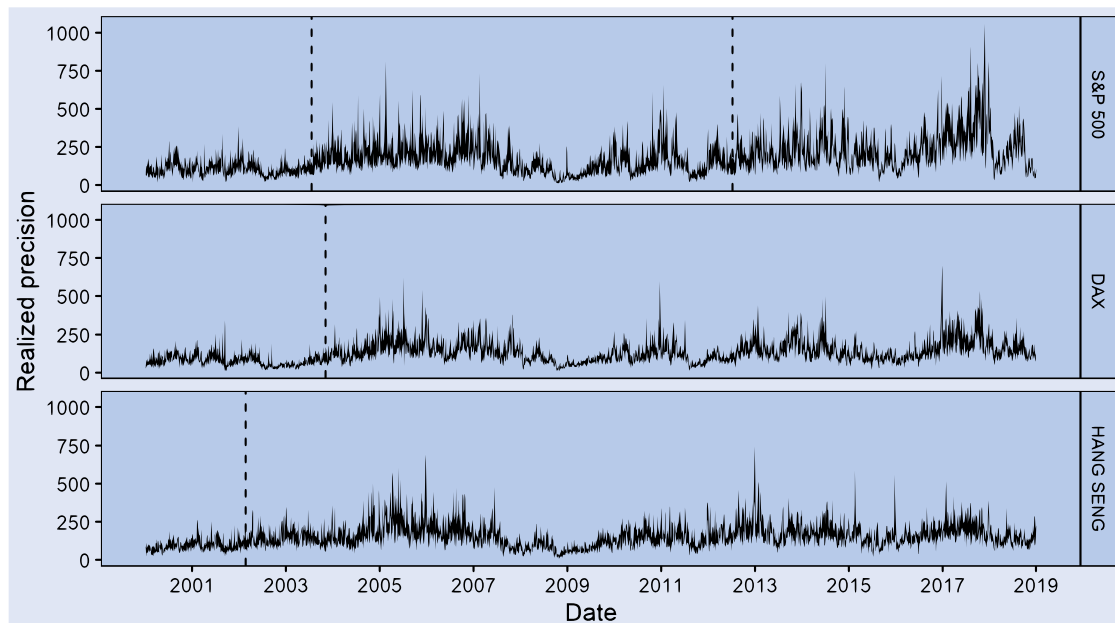


Figure 5. Realized precision and break-points for three stock market indices (2000–2018), realized kernel estimator. This figure shows plots of realized precision based on the realized kernel estimator of Barndorff-Nielsen *et al.* (2008) for the three stock market indices S&P 500 (first panel), DAX (second panel), and Hang Seng (third panel) from January 3, 2000, to December 31, 2018. Daily volatility estimates are based on the realized kernel in equation (12). Break-points are estimated using the two-step estimation procedure of Chan *et al.* (2014) with $k = 10$ and $p = 12$ and are highlighted by vertical dashed lines.

realized volatility series based on MedRV do not necessarily contain fewer estimated structural breaks than the series based on the realized kernel estimator. Again, we note that the unusually small value of $\sqrt{\text{MedRV}_t}$ for the Shanghai Composite on February 11, 2003, has a pronounced impact on the empirical results for the number of detected structural breaks (one or two breaks, depending on the lag length of the AR model) for log-realized volatility. In that regard, the results for the Shanghai Composite are distinctly different from the

other stock market indices. Thus, while one might reasonably assume $\sqrt{\text{MedRV}_t}$ to mitigate the problem of estimating potentially spurious structural breaks that are induced by price jumps, the empirical results do not unambiguously point into this direction.[†]

[†] Besides the realized volatility of stock market indices, we have also considered the (transformed) realized volatility based on the MedRV

This empirical finding is noteworthy since the log-transformation, providing favorable finite sample properties and avoiding the problem of imposing parameter restrictions to ensure positive values in estimation and forecasting, changes the linear dynamics of the realized volatility series fundamentally. As such, log-realized volatility follows a different process than realized volatility with significant implications for the detection of structural breaks in the underlying stochastic process. While the reciprocal of realized volatility also exhibits different autoregressive dynamics, compared to the original series, the differences are not as extreme. In the case of the log-transformation, the reduction of extreme values in the dependence on past values, as depicted exemplary in figure 1, seems to have a significant impact on the detection of breaks. Such extreme values can change the persistence of a series and may thus be detected as structural breaks in the autoregressive dynamics. As a result, an approach to structural break detection in the autoregressive dynamics of a series such as the one by Chan *et al.* (2014) is prone to react differently across the (un-)transformed series. Lastly, note that a change in the persistence of the series due to stale intraday prices in the calculation of the realized volatility measure is not likely to pose a problem in this application since only highly liquid indices are considered.

However, there is some evidence that in the HAR-framework, considering log-realized measures, break-point models can be well-specified when accounting for breaks in the variance (see the sparse Bayesian change-point model of Dufays and Rombouts 2019, where the authors, for example, find four breaks for the S&P 500 and five breaks for the DAX during the time from January 3, 2000, to August 5, 2015). Clearly, such a setting is not addressed well by the procedure of Chan *et al.* (2014), which only detects breaks following a change in the variance of the underlying segmented locally stationary AR model if this change leads to a detectable change in persistence. Thus, the effects that Box-Cox transformations of realized volatility have on the detection of structural breaks in the underlying volatility process do not seem clear-cut, even in a locally linear setting as considered here and in Liu and Maheu (2008) and Dufays and Rombouts (2019). While the existence of structural breaks in the volatility process is reasonable from an economic point of view and our understanding of these breaks is essential for financial applications, the question which breaks may be deemed true breaks and how to detect them best is a difficult one to answer. Therefore, we now compare the above results with results obtained from a more general estimation setup.

estimator for a sample of 19 liquid stocks. Considering individual-level stocks constitutes an interesting empirical exercise in its own right since the idiosyncratic features of these stocks or specific trading days may have an impact on the persistence of the realized volatility series. However, also considering individual-level stocks, we find that results are in line with the reported results for the (transformed) realized volatility of stock market indices. These additional results are available upon request.

4.2. Structural breaks detected in a linear regression framework

As a robustness check, we cast the above problem into the framework of testing for changes in the coefficients within a linear regression model.[†] Given the asymptotic theory outlined in Bai and Perron (1998) and the practical algorithm to estimate break-points in Bai and Perron (2003), such a setting may be a reasonable choice in cases where we do not expect too many breaks, with a priori unknown number and timing, occurring in a linear model (for applications to log-realized volatility, see Choi *et al.* 2010, Jung and Maderitsch 2014). Following Bai and Perron (2003) and since we are primarily interested in the number and timing of structural breaks, we consider the $m + 1$ regimes model without partial structural breaks:

$$Y_t = c_j + \beta_j' X_t + \varepsilon_t, \quad (13)$$

where $t = t_{j-1} + 1, \dots, t_j$, $j = 1, \dots, m + 1$, c_j is a constant, and X_t contains lagged values of Y_t .[‡]

Results for a range of lagged values in line with the results previously reported for the two-step Lasso procedure are depicted in tables 5 and 6 for the measures based on MedRV and the realized kernel estimator, respectively. On the one hand, we still find differences in the number of detected structural breaks between the considered Box-Cox transforms of realized volatility. On the other hand, it is evident that the break-point test of Bai and Perron (1998) delivers slightly different empirical results: Firstly, for smaller p , the number of detected breaks is often larger compared to the two-step Lasso procedure across all (un-)transformed series. The reverse is true for most series and a large p . Secondly, while we find zero breaks for larger p when considering the log-transforms of realized volatility, we also find zero breaks for some p and indices for untransformed realized volatility as well as for realized precision.

On a different note, if we only take into account structural breaks in the constant while ignoring dynamics (i.e. $\beta_j = \mathbf{0}$ in equation (13)), the results are relatively unaffected by the Box-Cox transform that is considered but we now find up to seven breaks. The latter finding is consistent with Jung and

[†] See also Casini and Perron (2019a) and Casini and Perron (2019b) for recent additions to the literature covering methods for the detection of structural breaks in continuous time.

[‡] The estimation can easily be implemented along the lines of Bai and Perron 2003 with the **strucchange** package in the statistical software environment R. See <https://CRAN.R-project.org/package=strucchange> and also Zeileis *et al.* (2003) for further comments. The parameter h , which governs the maximum number of structural breaks m that are allowed, is set to $h = 0.1$ when considering a model with breaks only in c_j (e.g. Jung and Maderitsch 2014) and we allow for a larger m by setting $h = 0.02$ when considering breaks in c_j and the dynamic effects β_j . The latter choice is based on the results reported for the two-step Lasso procedure where we find that structural breaks do not occur in quick succession. The Bayesian Information Criterion (BIC) is used to select the number of structural breaks. Note that the procedure of Chan *et al.* (2014) is computationally more efficient compared to the procedure outlined in Bai and Perron (2003), which is especially evident when considering long time series and several specifications. Thus, we have not considered a smaller number for h in our empirical application.

Table 5. Number of structural breaks for realized volatility series based on MedRV estimator, Bai and Perron (2003).

Index	Mean	AR(2)	AR(4)	AR(6)	AR(8)	AR(10)	AR(12)	AR(14)	AR(16)	AR(18)	AR(20)	AR(22)
Panel A: realized volatility												
DJIA	5	5	3	2	2	2	2	2	2	2	2	2
CAC 40	6	10	9	4	4	3	0	0	0	0	0	0
FTSE 100	6	9	7	4	3	2	2	2	2	2	2	2
DAX	7	10	6	4	3	2	2	2	2	2	0	0
Hang Seng	7	8	5	3	2	2	2	0	0	0	0	0
Nikkei 225	6	4	3	3	2	2	2	0	0	0	0	0
S&P 500	4	5	3	3	2	2	2	2	2	2	2	2
Shanghai Composite	6	10	4	4	0	0	0	0	0	0	0	0
Panel B: log-realized volatility												
DJIA	6	5	0	0	0	0	0	0	0	0	0	0
CAC 40	7	5	0	0	0	0	0	0	0	0	0	0
FTSE 100	6	5	0	0	0	0	0	0	0	0	0	0
DAX	7	7	0	0	0	0	0	0	0	0	0	0
Hang Seng	7	5	3	0	0	0	0	0	0	0	0	0
Nikkei 225	6	4	0	0	0	0	0	0	0	0	0	0
S&P 500	6	5	2	0	0	0	0	0	0	0	0	0
Shanghai Composite	6	9	2	0	0	0	0	0	0	0	0	0
Panel C: realized precision												
DJIA	7	6	3	0	0	0	0	0	0	0	0	0
CAC 40	7	10	10	10	10	7	4	0	0	0	0	0
FTSE 100	6	6	3	3	3	2	2	2	0	0	0	0
DAX	7	8	0	0	0	0	0	0	0	0	0	0
Hang Seng	7	6	3	2	2	0	0	0	0	0	0	0
Nikkei 225	7	6	0	0	0	0	0	0	0	0	0	0
S&P 500	6	6	5	0	0	0	0	0	0	0	0	0
Shanghai Composite	1	2	2	2	2	2	2	2	2	2	2	2

Notes: This table lists the estimated number of break-points for all eight stock market indices and the realized volatility (Panel A), log-realized volatility (Panel B), and realized precision (Panel C) time series based on the MedRV estimator of Andersen *et al.* (2012) for breaks in the mean (Mean) of the respective series and across different AR specifications. The procedure of Bai and Perron (2003) is used to estimate the number of structural breaks. The upper bound for the number of breaks is governed by the parameter $h = 0.1$ for breaks in the mean and $h = 0.02$ for breaks in the dynamics, respectively.

Table 6. Number of structural breaks for realized volatility series based on realized kernel estimator, Bai and Perron (2003).

Index	Mean	AR(2)	AR(4)	AR(6)	AR(8)	AR(10)	AR(12)	AR(14)	AR(16)	AR(18)	AR(20)	AR(22)
Panel A: realized volatility												
DJIA	6	10	3	2	2	0	0	0	0	0	0	0
CAC 40	6	10	6	0	0	0	0	0	0	0	0	0
FTSE 100	6	8	4	2	2	2	2	2	2	2	2	2
DAX	7	10	7	4	4	2	2	1	0	0	0	0
Hang Seng	7	7	5	3	2	2	2	2	2	0	0	0
Nikkei 225	6	8	5	4	4	4	4	4	4	4	4	2
S&P 500	5	10	3	3	2	2	2	2	2	2	2	0
Shanghai Composite	6	8	4	4	0	0	0	0	0	0	0	0
Panel B: log-realized volatility												
DJIA	6	7	1	1	0	0	0	0	0	0	0	0
CAC 40	7	8	3	0	0	0	0	0	0	0	0	0
FTSE 100	6	6	0	0	0	0	0	0	0	0	0	0
DAX	7	10	1	1	0	0	0	0	0	0	0	0
Hang Seng	7	9	4	0	0	0	0	0	0	0	0	0
Nikkei 225	6	7	1	0	0	0	0	0	0	0	0	0
S&P 500	7	9	3	1	1	1	0	0	0	0	0	0
Shanghai Composite	6	8	2	0	0	0	0	0	0	0	0	0
Panel C: realized precision												
DJIA	7	9	4	3	3	0	0	0	0	0	0	0
CAC 40	7	8	3	0	0	0	0	0	0	0	0	0
FTSE 100	6	10	2	2	0	0	0	0	0	0	0	0
DAX	7	8	4	1	0	0	0	0	0	0	0	0
Hang Seng	7	10	4	0	0	0	0	0	0	0	0	0
Nikkei 225	6	9	1	0	0	0	0	0	0	0	0	0
S&P 500	7	7	5	1	1	0	0	0	0	0	0	0
Shanghai Composite	7	8	4	2	0	0	0	0	0	0	0	0

Maderitsch (2014), who detect up to six breaks in the mean of log-realized volatility for three stock market indices using the procedure of Bai and Perron (2003). Overall, these results point towards the presence of structural breaks in the mean of Box-Cox transforms of realized volatility but not necessarily in the dynamics of the series. This finding and the results reported in Choi *et al.* (2010) and Jung and Maderitsch (2014) are not in line with the HAR setting studied by Liu and Maheu (2008) and Dufays and Rombouts (2019), where no change in the constant of the HAR model is detected. In order to shed some light on the contradicting results for the two-step group Lasso procedure, a simple Monte Carlo study is entertained in the next section.

5. Limits of Lasso? A simple Monte Carlo study

We turn again to the potential impact of price jumps on the detection of structural breaks. As mentioned above and illustrated in figure 1, the log-transformation reduces the number of extreme values in the dependence on past values. Thus, in this case, price jumps are likely to have a less significant impact on the detection of break-points. The behavior of the two-step group Lasso procedure in the presence of price jumps with or without log-transformation is further investigated in the framework of a stochastic volatility jump diffusion model. This model has been studied widely and is also employed by, among others, Huang and Tauchen (2005). It reads as follows:

$$\begin{aligned} dp_t &= \mu dt + \exp[\beta_0 + \beta_1 v_t] dW_{p,t} + dJ_t, \\ dv_t &= \alpha_v v_t dt + dW_{v,t}, \end{aligned} \quad (14)$$

where $W_{p,t}$ and $W_{v,t}$ are standard Brownian motions with leverage correlation ρ , v_t is a stochastic volatility factor, and J_t is a compound Poisson process with constant jump intensity λ and jump size distributed as $\mathcal{N}(0, \sigma_J^2)$. Assuming a trading day of 6.5 hours, prices on a 5-minute frequency are sampled and, subsequently, 5-minute returns are computed. From this, the (log-)realized volatility for each trading day is obtained. Note that we restrict the simulations to these two measures since (i) they are actually used in empirical research and (ii) the differences in results reported for these measures are most striking.

As a special case, we also consider a model without a jump term in the price process:

$$\begin{aligned} dp_t &= \mu dt + \exp[\beta_0 + \beta_1 v_t] dW_{p,t}, \\ dv_t &= \alpha_v v_t dt + dW_{v,t}. \end{aligned} \quad (15)$$

In the following, results are shown for three sample sizes, namely $n \in \{1024, 2048, 4096\}$. In line with Huang and Tauchen (2005), we fix $\mu = 0.3$, $\beta_0 = 0$, $\beta_1 = 0.125$, and $\rho = -0.62$. The jump intensity is set to $\lambda = 0.014$ and we vary the variance of the jump size σ_J^2 and allow for $\sigma_J^2 \in \{0.5, 1, 1.5\}$. In our simulation experiments, we consider two different scenarios, i.e. $\alpha_v = -0.1$ (medium mean reversion) and $\alpha_v = -1.386$ (fast mean reversion). Simulations are based on 1000 replications and we set the maximum lag length

in the two-step group Lasso procedure to $p = 4$ throughout the simulation experiments. As it turns out, longer lags are often not statistically significant across replications which is why a smaller value for p should suffice.

Before we introduce structural breaks into this framework, we first investigate the performance for the case of zero structural breaks both in a model with and without jump term, i.e. the models given by equations (14) and (15). Here, only the case of $n = 4096$ is discussed exemplarily. Since we know ex ante that there are zero structural breaks in the DGP, we lower the maximum number of breaks in the first step of the two-step group Lasso procedure to $k = 5$. Results can be found in Panel A of table 7. As can be seen, the percentage of correctly estimated (pce) number of break-points is close to one for both realized and log-realized volatility when there are no jumps (i.e. $\lambda = 0$) or the variance of the jump size is rather low (i.e. $\sigma_J^2 \in \{0.5, 1\}$). In general, there are more cases where the two-step group Lasso procedure falsely detects structural breaks for realized volatility than for log-realized volatility. In particular for $\sigma_J^2 = 1.5$ and a fast mean reversion in the volatility process. Thus, the log-transformation effectively reduces the magnitude of the price jumps across different values of σ_J^2 and leads to better results in the case of zero structural breaks in the DGP.

In order to assess the impact of price jumps on the detection of structural breaks, we now introduce two breaks into the model given by equation (14). The break-fractions are set to $\tau_1 = 0.5$ and $\tau_2 = 0.75$. The parameterization of the DGP changes with α_v , where we consider the above-mentioned scenarios, and is given by:

$$\begin{aligned} \alpha_v &= -1.386 \text{ (fast mean reversion),} & \text{if } t \leq t_1, \\ \alpha_v &= -0.1 \text{ (medium mean reversion),} & \text{if } t_1 < t \leq t_2, \\ \alpha_v &= -1.386 \text{ (fast mean reversion),} & \text{if } t > t_2. \end{aligned} \quad (16)$$

Similar to the empirical application above, we set the maximum number of breaks in the first step of the two-step group Lasso procedure to $k = 10$. Results can be found in Panel B of table 7. Based on the pce, the detection of the two break-points works better for log-realized volatility than for realized volatility across all values of σ_J^2 . Interestingly, the pce decreases again for both volatility series as σ_J^2 increases, albeit this decrease is way more noticeable for realized volatility. For example, considering $n = 2,048$ and $\sigma_J^2 = 0.5$, the pce for realized volatility is 0.882 and for log-realized volatility 0.944. For $n = 2,048$ and $\sigma_J^2 = 1.5$, the pce for realized volatility decreases to 0.489 and for log-realized volatility only to 0.871. If the number of break-points is estimated correctly, the estimated timing of the breaks is relatively accurate as can be seen from the estimated break-fractions. Lastly, a larger sample size leads to a slight decrease of the pce for both volatility series since these series potentially contain more price jumps with large jump size.

Overall, the simulation results are in line with the empirical finding illustrated in figure 1. The log-transformation seems to effectively reduce the impact of price jumps and thus leads to a better performance when estimating the number of break-points with the two-step group Lasso procedure of Chan *et al.* (2014). These price jumps are sometimes falsely taken

Table 7. Estimated break-points and break-fractions.

Simulated series	λ	σ_j^2	n	pce	$\hat{\tau}_1$	$\hat{\tau}_2$
Panel A: Zero structural breaks in DGP						
$\alpha_v = -0.1$ (medium mean reversion)						
$\sqrt{RV_t}$	0	–	4096	0.957	–	–
$\log(\sqrt{RV_t})$	0	–	4096	1.000	–	–
$\sqrt{RV_t}$	0.014	0.500	4096	0.971	–	–
$\log(\sqrt{RV_t})$	0.014	0.500	4096	1.000	–	–
$\sqrt{RV_t}$	0.014	1.000	4096	0.924	–	–
$\log(\sqrt{RV_t})$	0.014	1.000	4096	0.996	–	–
$\sqrt{RV_t}$	0.014	1.500	4096	0.650	–	–
$\log(\sqrt{RV_t})$	0.014	1.500	4096	0.956	–	–
$\alpha_v = -1.386$ (fast mean reversion)						
$\sqrt{RV_t}$	0	–	4096	1.000	–	–
$\log(\sqrt{RV_t})$	0	–	4096	1.000	–	–
$\sqrt{RV_t}$	0.014	0.500	4096	1.000	–	–
$\log(\sqrt{RV_t})$	0.014	0.500	4096	1.000	–	–
$\sqrt{RV_t}$	0.014	1.000	4096	0.916	–	–
$\log(\sqrt{RV_t})$	0.014	1.000	4096	1.000	–	–
$\sqrt{RV_t}$	0.014	1.500	4096	0.417	–	–
$\log(\sqrt{RV_t})$	0.014	1.500	4096	0.964	–	–
Panel B: Two structural breaks in DGP						
$\sqrt{RV_t}$	0.014	0.500	1024	0.857	0.500	0.733
$\sqrt{RV_t}$	0.014	0.500	2048	0.882	0.502	0.731
$\sqrt{RV_t}$	0.014	0.500	4096	0.793	0.504	0.728
$\log(\sqrt{RV_t})$	0.014	0.500	1024	0.885	0.497	0.739
$\log(\sqrt{RV_t})$	0.014	0.500	2048	0.944	0.498	0.739
$\log(\sqrt{RV_t})$	0.014	0.500	4096	0.920	0.498	0.740
$\sqrt{RV_t}$	0.014	1.000	1024	0.757	0.498	0.732
$\sqrt{RV_t}$	0.014	1.000	2048	0.797	0.499	0.731
$\sqrt{RV_t}$	0.014	1.000	4096	0.732	0.502	0.727
$\log(\sqrt{RV_t})$	0.014	1.000	1024	0.861	0.497	0.740
$\log(\sqrt{RV_t})$	0.014	1.000	2048	0.933	0.499	0.740
$\log(\sqrt{RV_t})$	0.014	1.000	4096	0.920	0.498	0.740
$\sqrt{RV_t}$	0.014	1.500	1024	0.470	0.479	0.720
$\sqrt{RV_t}$	0.014	1.500	2048	0.428	0.489	0.723
$\sqrt{RV_t}$	0.014	1.500	4096	0.364	0.484	0.711
$\log(\sqrt{RV_t})$	0.014	1.500	1024	0.798	0.495	0.740
$\log(\sqrt{RV_t})$	0.014	1.500	2048	0.871	0.498	0.740
$\log(\sqrt{RV_t})$	0.014	1.500	4096	0.882	0.497	0.740

Notes: This table depicts, for different sample sizes n , the percentage of correctly estimated (pce) number of break-points as well as the average break-fractions $\hat{\tau}_1$ and $\hat{\tau}_2$ across 1,000 replications based on the models in equations (14) to (16) for both realized and log-realized volatility ($\sqrt{RV_t}$ and $\log(\sqrt{RV_t})$, respectively). The jump intensity (λ) is fixed in the case of active jumps and the variance of the jump size (σ_j^2) varies. Panel A illustrates results for the case of zero breaks, Panel B for the case of two breaks.

to be structural breaks when realized volatility is not transformed. Thus, the above-mentioned empirical findings based on log-realized volatility are likely to be a closer approximation to reality and the log-transformation is a reasonable choice in empirical applications involving realized volatility.

6. Conclusion

Given that financial markets change over time, structural breaks should be considered in our approaches to the modeling and forecasting of realized volatility. By using two Box-Cox transformations of realized volatility and a recently developed two-step group Lasso procedure to consistently estimate structural breaks in the autoregressive dynamics of a series, we show that transformations of realized volatility significantly alter the dynamics of the volatility series and with it

the number of breaks that are detected in such a locally linear setting. The results are especially striking for log-realized volatility, for which no structural breaks are detected across a variety of autoregressive specifications. It thus seems to be highly relevant to factor in the effects of this often-used (non-linear) transformation when applying break-point detection procedures. Doing so is likely to help answer the question *if* there are structural breaks in the underlying volatility process and, ultimately, which of these may be deemed true structural breaks.

Importantly, in a Monte Carlo study based on a stochastic volatility jump diffusion model, we find that jumps significantly affect the detection of structural breaks via the two-step group Lasso procedure in the dynamics of realized volatility time series. The log-transformation reduces the impact of these jumps and thus seems to be a reasonable choice in empirical applications. These findings would suggest that there are likely zero or only very few structural breaks in the

dynamics of most of the realized volatility series considered in our empirical application.

While this study presents some interesting and robust empirical findings, further research is necessary to better understand the different results reported in the empirical literature. Moreover, from a practical point of view, this could also be important in the context of volatility forecasting. As research related to local-to-zero breaks by, for example, Boot and Pick (2020) and Pesaran and Timmermann (2005) has shown, not every structural break necessarily has a significant impact in forecasting exercises. This diminished effect of some structural breaks in forecasts may also be affected by Box-Cox transforms of the variable of interest. A thorough investigation of such an effect would have to reconcile the findings of the former with results reported by, for example, Lütkepohl and Xu (2012) and Taylor (2017).

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